

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T39 · Standard and Compound Units

Questions

Student book · questions with writing space

■ 16 pages

FAMILY MAP

Standard Compound Units

F01	Use pressure as a compound measure	1 Q 3 marks
F02	Use speed, distance and time	10 Q 30 marks
F03	Select and convert standard metric measures	1 Q 2 marks
F04	Use density as a compound measure	1 Q 2 marks
F05	Read time and measuring instruments	1 Q 6 marks

Question 05

4MA1-1H Q8

MODULAR UNIT 1

3 marks

- 8 A cylinder is placed on a table.

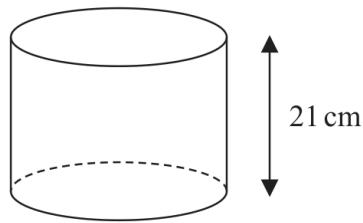


Diagram **NOT**
accurately drawn

The volume of the cylinder is 1575 cm^3

The force exerted by the cylinder on the table is 84 newtons.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the pressure on the table due to the cylinder.

..... newtons/cm²

4MA1-2H Q3

3 marks

- Work out the total distance the plane flies.

.....km

Question 20

4MA1-2H Q6

MODULAR UNIT 1

3 marks

- 6 Change a speed of w metres per second to a speed in kilometres per hour.
Give your answer in terms of w in its simplest form.

..... kilometres per hour

Question 16

4MA1-2H Q9

MODULAR UNIT 1

3 marks

- 9 Change a speed of 27 kilometres per hour to a speed in metres per second.

..... m/s

Question 16

4MA1-2H Q3 Q3

MODULAR UNIT 1

3 marks

- 3 An aeroplane travelled from New York City to Los Angeles.

The aeroplane travelled a distance of 3980 kilometres in 5 hours 24 minutes.

Work out the average speed of the aeroplane.

Give your answer in kilometres per hour correct to the nearest whole number.

..... kilometres per hour

4MA1-2H Q7 Q7

3 marks

..... metres per second

- 9 The density of gold is 19.3 g/cm^3
A gold bar has volume 150 cm^3

Work out the mass of the gold bar.

..... g

..... kilometres per hour

- Work out the average speed of the train.
Give your answer in km/h correct to one decimal place.

..... km/h

The distance the plane flew was 5658 km.
The flight time was 8 hours 12 minutes.

..... km/h

Give your answer in seconds, correct to the nearest second.

(Total for Question 6 is 3 marks)

- 2 Change a speed of 90 kilometres per hour to a speed in metres per second.
Show your working clearly.

..... m/s

(Total for Question 2 is 3 marks)

..... cm³

A diagram of a cone. A vertical line segment from the apex to the center of the base is labeled $5x \text{ cm}$. A horizontal line segment from the center of the base to the edge is labeled $x \text{ cm}$. The base is represented by a dashed line for the hidden part and a solid line for the visible part.

Work out the density of the hemisphere.
Show your working clearly.

..... g/cm³

(Total for Question 26 is 6 marks)